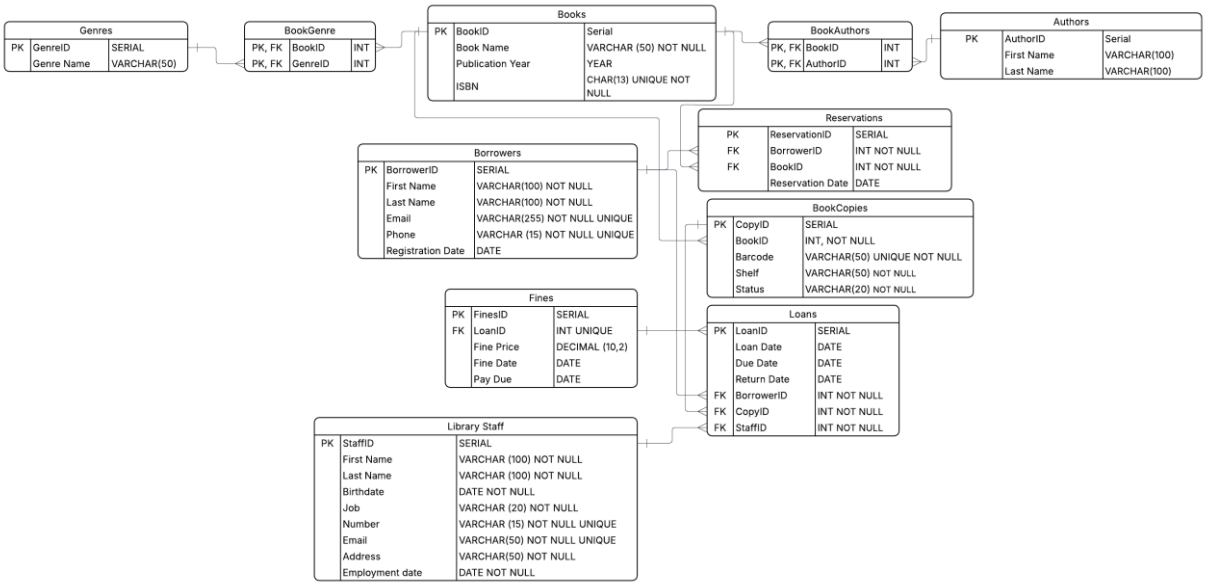
1.1 Logical Scheme



1.2 Objects

Books

This table stores the core information for every book title in the library catalog.

Table Name	Field Name	Field Description	Data Type / Constraints
Books	BookID	Unique indentifier for the book title	SERIAL, PK
	Book Name	Title of the book	VARCHAR(50), NOT NULL
	Publication Year	The year the book was released	YEAR
	ISBN	Numeric commercial book identifier	CHAR(13) UNIQUE NOT NULL

One to many with **BookCopies**: one book title can have multiple physical copies

Many to many with **Authors** and **Genres** using bridge tables

Example with data

BookID	Book Name	Publication Year	ISBN
1	The Great Gatsby	1925	9780743273565

Authors

This table stores information about the creators of the books in the library system.

Table Name	Field Name	Field Description	Data Type / Constraints
Authors	AuthorID	Unique identifier for the author	SERIAL, PK
	First Name	Author's first name	VARCHAR(50), NOT NULL
	Last Name	Author's last name	VARCHAR(50), NOT NULL

Many to many with **Books** through the **BookAuthors** bridge table: one author can write many books, and one book can have multiple authors

Example with data

AuthorID	First Name	Last Name
105	Scott	Fitzgerald

BookAuthors

This table resolves the many to many relationship between book and authors.

Table Name	Field Name	Field Description	Data Type / Constraints
BookAuthors	BookID	Reference to the book	INT, PK, FK
	AuthorID	Reference to the author	INT, PK, FK

Uses a composite primary key to ensure a specific author isn't linked to the same book twice

Example with data

BookID	AuthorID
1	105

Genres

A lookup table, like a dictionary, of all literary categories available in the library.

Table Name	Field Name	Field Description	Data Type / Constraints
Genres	GenreID	Unique identifier for the genre	SERIAL, PK
	Genre Name	The name of the category	VARCHAR(50), NOT NULL, UNIQUE

Many to many with **Books** through the **BookGenre** bridge table

Example with data

GenreID	Genre Name
5	Science Fiction
6	Classic Literature

BookGenre

This junction table resolves the many to many relationship between books and their literary genres

Table Name	Field Name	Field Description	Data Type / Constraints
BookGenre	BookID	Reference to the book title	INT, PK, FK
	GenreID	Reference to the genre category	INT, PK, FK

Also uses a composite primary key to ensure a book is never assigned to the same genre more than once

Example with data

BookID	GenreID

1	6

Library Staff

This table contains comprehensive profiles for library employees.

Table Name	Field Name	Field Description	Data Type / Constraints
Library Staff	StaffID	Unique identifier for the staff member	SERIAL, PK
	First Name	Staff member's first name	VARCHAR(100), NOT NULL
	Last Name	Staff member's last name	VARCHAR(100), NOT NULL
	Birthdate	Employee's date of birth	DATE, NOT NULL
	Job	Specific job title or role	VARCHAR(20), NOT NULL
	Number	Contact phone number	VARCHAR(15), NOT NULL, UNIQUE
	Email	Professional email address	VARCHAR(50), NOT NULL, UNIQUE
	Address	Residential address	VARCHAR(50), NOT NULL
	Employment date	Date the staff member started working	DATE, NOT NULL

One to many with **Loans** - one staff member can be responsible for authorizing and processing many different loan transactions.

Example with data

StaffID	First Name	Last Name	Birthdate	Job	Number	Email	Address	Employment date
50	Jane	Smith	1985-05-15	Librarian	+77777777777	j.smith@gmail.com	123 Maple St	2026-01-10

Borrowers

This table stores the information and contact details for all registered library members(borrowers)

Table Name	Field Name	Field Description	Data Type / Constraints
Borrowers	BorrowerID	Unique identifier for the member	SERIAL, PK
	First Name	Member's first name	VARCHAR(100), NOT NULL
	Last Name	Member's last name	VARCHAR(100), NOT NULL
	Email	Professional email address	VARCHAR(255), NOT NULL, UNIQUE
	Number	Contact phone number	VARCHAR(15), NOT NULL, UNIQUE
	Registration Date	Date the member joined	DATE, NOT NULL

One to many with **Reservations** - a borrower can place multiple holds on books

One to many with **Loans** - a borrower can check out multiple books over time

Example with data

BorrowerID	First Name	Last Name	Email	Number	Registration date
10	John	Doe	j.doe@gmail.com	+77777777776	2026-03-13

BookCopies

This table tracks the individual physical items owned by the library

Table Name	Field Name	Field Description	Data Type / Constraints
BookCopies	CopyID	Unique identifier for the physical unit	SERIAL, PK
	BookID	Reference to the book title	INT, FK, NOT NULL
	Barcode	Scannable code for the book	VARCHAR(50), UNIQUE, NOT NULL
	Shelf	Physical location in the library	VARCHAR(50), NOT NULL
	Status	Book availability	VARCHAR(20), NOT NULL

Many to one with **Books** - connects a specific copy to its general title information

One to many with **Loans** - tracks which specific copy was taken by a borrower

Example with data

CopyID	BookID	Barcode	Shelf	Status
1001	1	BC1001GATSBY	A-01-01	Available

Loans

This table is for library transactions, recording when a book copy is checked out and returned

Table Name	Field Name	Field Description	Data Type / Constraints
Loans	LoanID	Unique identifier for the transaction	SERIAL, PK
	Loan Date	Date the book was borrowed	DATE, NOT NULL
	Due Date	Date the book is due for return	DATE, NOT NULL
	Return Date	Actual date the book was return	DATE
	BorrowerID	Reference to the member	INT, FK, NOT NULL
	CopyID	Reference to the physical copy	INT, FK, NOT NULL
	StaffID	Reference to the staff member	INT, FK, NOT NULL

Acts as a junction connecting **Borrowers, BookCopies** and **Library staff**

Parent table to **Fines**.

Example with data

LoanID	Loan Date	Due Date	Return Date	BorrowerID	CopyID	StaffID
501	2026-01-01	2026-01-15	2026-01-14	10	1001	50

Fines

This table tracks financial penalties for overdue loans.

Table Name	Field Name	Field Description	Data Type / Constraints
Fines	FinesID	Unique identifier for the fine	SERIAL, PK
	LoanID	Reference to the specific loan	INT, FK, UNIQUE
	Fine Price	The monetary amount of the fine	DECIMAL(10,2)
	Fine Date	Date the fine was issued	DATE
	Pay Due	The deadline for the fine payment	DATE

This

Example with data

FinesID	LoanID	Fine Price	Fine Date	Pay Due
1	501	15.00	2026-01-09	2026-01-24

Reservations

This table manages “holds” placed by borrowers on specific book titles.

Table Name	Field Name	Field Description	Data Type / Constraints
Reservations	ReservationID	Unique identifier for the hold	SERIAL, PK
	BorrowerID	Reference to the member waiting	INT, FK, NOT NULL
	BookID	Reference to the book title	INT, FK, NOT NULL
	Reservation Date	Date the reservation was made	DATE

Linked to **Borrowers** and **Books**. Unlike Loans, this links to the general BookID rather than a specific copy.

Example with data

ReservationID	BorrowerID	BookID	Reservation Date
101	10	1	2026-01-01